

Efficiency of a Carnot Engine at Maximum Power Output



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Course Presentation-Thermal Physics

Abstract

This presentation explores how efficiency of a Carnot's engine changes when power output is constrained by the rates of heat transfer to and from the working substance. The efficiency η is given by the expression

$$\eta = 1 - \left(\frac{T_2}{T_1}\right)^{1/2}$$

where T_1 and T_2 are the respective temperatures of the heat source and heat sink.

Power of a normie Carnot engine

We know real-world engines are not efficient as the classical Carnot engine. The corresponding efficiency has a limit due to entropy constraints, and during its calculation it is considered that the isothermal and adiabatic processes are infinitely slow, so the power output of an idealized Carnot's engine is zero.

$$P_{normie} = 0$$

$$\eta = 1 - \frac{T_2}{T_1}$$

Finite power engine

Quite surprised, huh? Can it operate at finite power? Well, if answer was no, then probability of this presentation taking place would be near zero!

The cycle is sped up so that the working substance is colder than the heat reservoir during expansion and correspondingly hotter than the sink during compression.

$$T_{1w} < T_1$$

$$T_{2w} > T_2$$

Finite power engine

If the two isothermal stages occur without any change in the temperature of the working substance, heat flows directly from source to sink, meaning $W = 0 \Rightarrow P = 0$ and $\eta = 0$ as well.

Somewhere between these two zero points (i.e. optimum and zero efficiency) lies the condition for maximum power output, for which we will calculate the efficiency.

Heat Flux

Heat flux is defined as flow of heat per unit area per unit time. It is vector quantity. Considering conduction across reservoirs, we can write flux as

$$\phi = k \frac{A}{L} (T_2 - T_1)$$

Furthermore, assuming constant reservoir parameters, we can write

$$\frac{dW}{dt} = k (T_2 - T_1)$$

Hence work done in isothermal processes are:

$$W_1 = \alpha t_1 (T_1 - T_{1w}) \text{ for isothermal I with time } t_1$$

$$W_1 = \beta t_2 (T_{2w} - T_1) \text{ for isothermal II with time } t_2$$

Entropy: The Ultimate Variable

The adiabatic processes then bring the working substance back to its original volume and temperature. Since the process is cyclic,

$$\Delta S_{cycle} = 0$$

hence by calculating entropy change for each process we get

$$\frac{W_1}{T_{1w}} = \frac{W_2}{T_{2w}}$$

$$\Rightarrow \left(\frac{t_1}{t_2}\right) = \frac{\beta T_{1w}(T_{2w} - T_2)}{\alpha T_{2w}(T_1 - T_{1w})}$$

Adiabatic got time rizzz

A natural question, what is the time taken to complete adiabatic process, given the time to complete corresponding isothermal process?

The work done in adiabatic process is

$$W_{adiabatic} = \frac{nR(T_2 - T_1)}{\gamma - 1}$$

Hence one can draw an analogy from work done in isothermal process, considering constant power that

$$t'_1 = \frac{t_1}{\gamma - 1}$$

Therefore total time taken to complete whole cycle is $\frac{\gamma}{\gamma-1}(t_1 + t_2)$

Time for calculations!

The power (P) of the engine is then given by the expression

$$P = \frac{(\gamma - 1)(W_1 - W_2)}{(t_1 + t_2)\gamma}$$

Eliminating t_1, t_2

$$P = (\gamma - 1)\alpha\beta xy (T_1 - T_2 - x - y) \frac{1}{\gamma[\beta T_1 y + \alpha T_2 x + xy(\alpha - \beta)]}$$

$$x = T_1 - T_{1w}, \quad y = T_{2w} - T_2$$

Condition of Maximum Power

P is maximized by values of x and y satisfying the following equations:

$$\left(\frac{\partial P}{\partial x}\right) = 0,$$

i.e.

$$\begin{aligned} & \beta T_1 y (T_1 - T_2 - x - y) \\ = & x [\beta T_1 y + \alpha T_2 x + xy(\alpha - \beta)] \end{aligned}$$

(1a)

$$\left(\frac{\partial P}{\partial y}\right) = 0$$

i.e.

$$\begin{aligned} & \alpha T_2 x (T_1 - T_2 - x - y) \\ = & y [\beta T_1 y + \alpha T_2 x + xy(\alpha - \beta)] \end{aligned}$$

(1b)

Hawk's dilemma of 2^a

From the definitions it follows that,

$$y = \left(\frac{\alpha T_2}{\beta T_1} \right)^{1/2} x \quad (2)$$

which is then used to eliminate y from Eq. (1a) to give $\frac{x}{T_1} = \mu$:

$$[1 - (\alpha/\beta)]\mu^2 - 2 \left[\left(\frac{\alpha T_2}{\beta T_1} \right)^{1/2} + 1 \right] \mu + \left[1 - \left(\frac{T_2}{T_1} \right) \right] = 0 \quad (3)$$

Hawk's dilemma of 2^a

Since $\mu < 1$, the physically relevant solution of this equation is readily shown to be:

$$\mu = \frac{x}{T_1} = \frac{1 - (T_2/T_1)^{1/2}}{1 + (\alpha/\beta)^{1/2}} \quad (4)$$

From Eq. (2) it follows that

$$\frac{y}{T_2} = \frac{(T_1/T_2)^{1/2} - 1}{1 + (\beta/\alpha)^{1/2}} \quad (5)$$

Finally the efficiency we all are desperate about

The efficiency of the engine (η') at maximum power is given by

$$\begin{aligned}\eta' &= (W_1 - W_2) / W_1 \\ &= 1 - (T_{2w} / T_{1w}) \\ &= 1 - [(T_2 + y) / (T_1 - x)]\end{aligned}\tag{6}$$

Using Eqs. (4) and (5) to eliminate x and y for η' leads to the result

$$\boxed{\eta' = 1 - (T_2 / T_1)^{1/2}}\tag{7}$$

References

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